

Model Tuning Summary

Model	What was tuned	Best settings found	Why it was tried
Linear Regression	No tuning (baseline)	N/A	Simplest possible baseline — a straight-line relationship between features and claim
Polynomial Regression	Degree (2 vs. 3)	Degree 2 (degree 3 overfit — CV R^2 dropped to 0.27)	Tests whether a curved, rather than straight, relationship fits better
Random Forest	Tree count, depth, split rules	200 trees, max depth 10, min samples split 5, min samples leaf 2	Can capture interactions and non-linear effects automatically
Support Vector Regression (SVR)	Kernel, regularization (C), margin (epsilon)	Linear kernel, C=10, epsilon=0.5	A different mathematical approach, useful as a sanity check against the others
XGBoost	Tree count, depth, learning rate, subsampling	100 trees, max depth 3, learning rate 0.05, subsample 1.0	Often the strongest performer on structured, tabular data like this